SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I VAISHNAVI.S 192521313 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

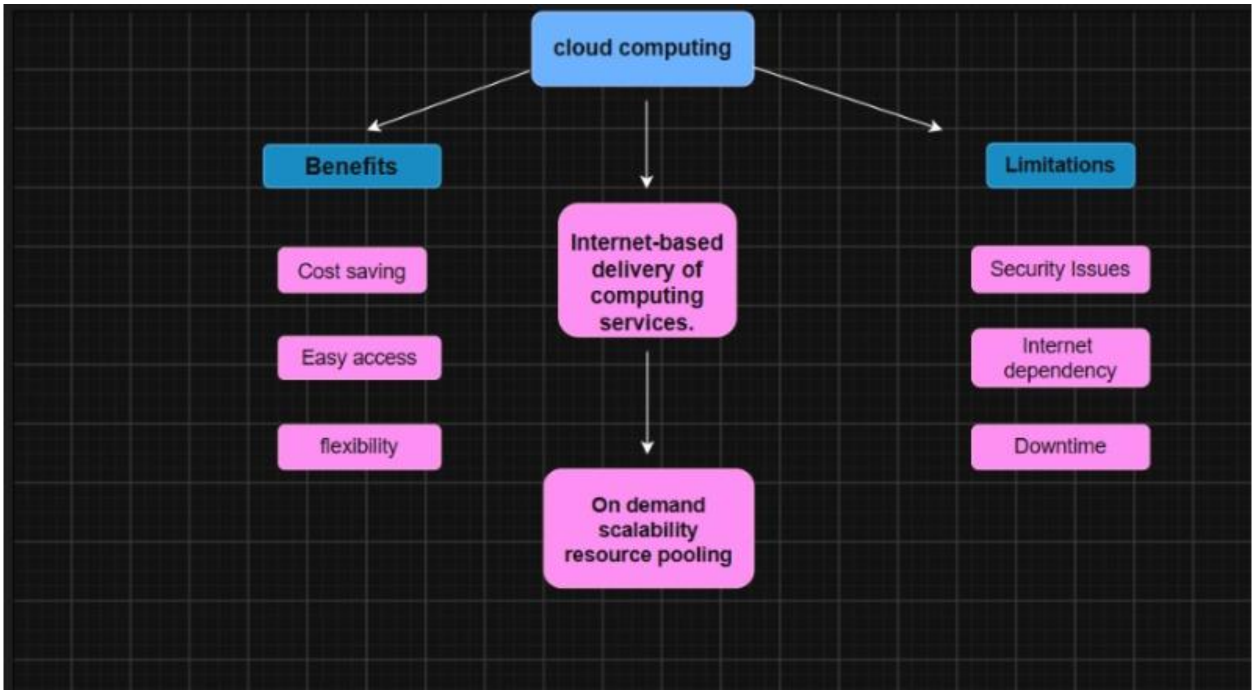
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

5. Questions

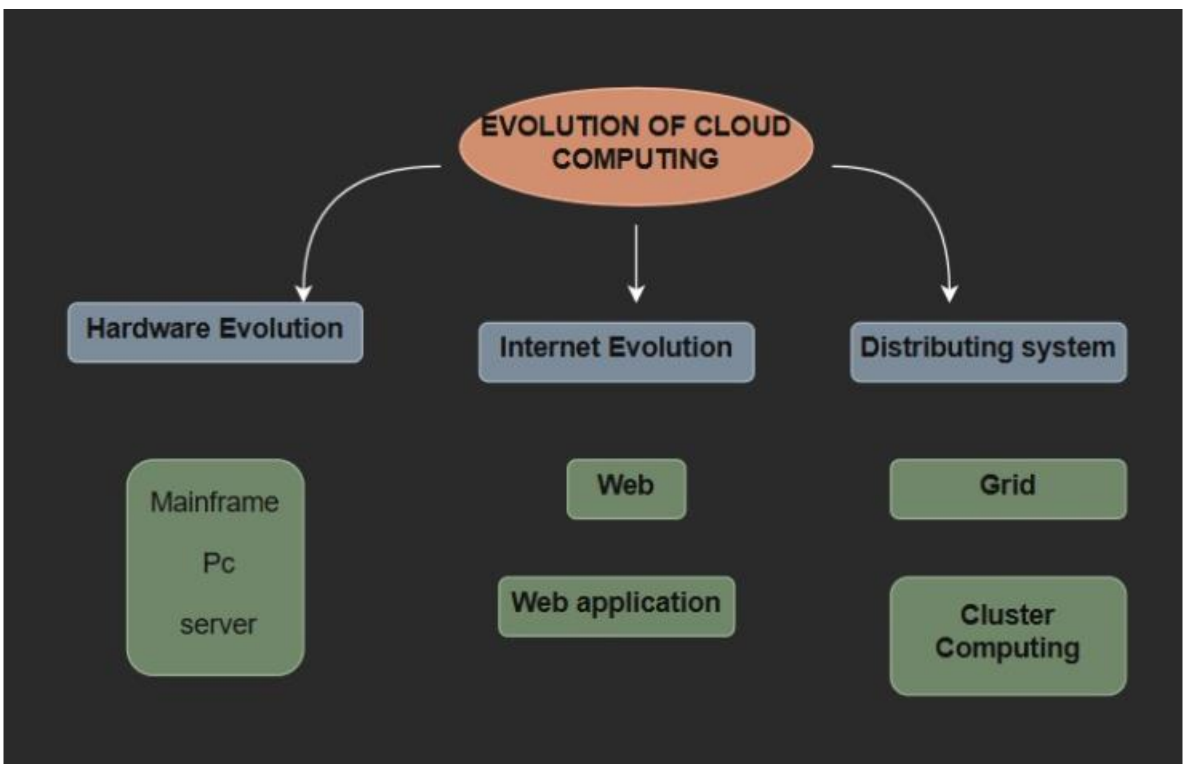
i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Concept Mapping Tasks

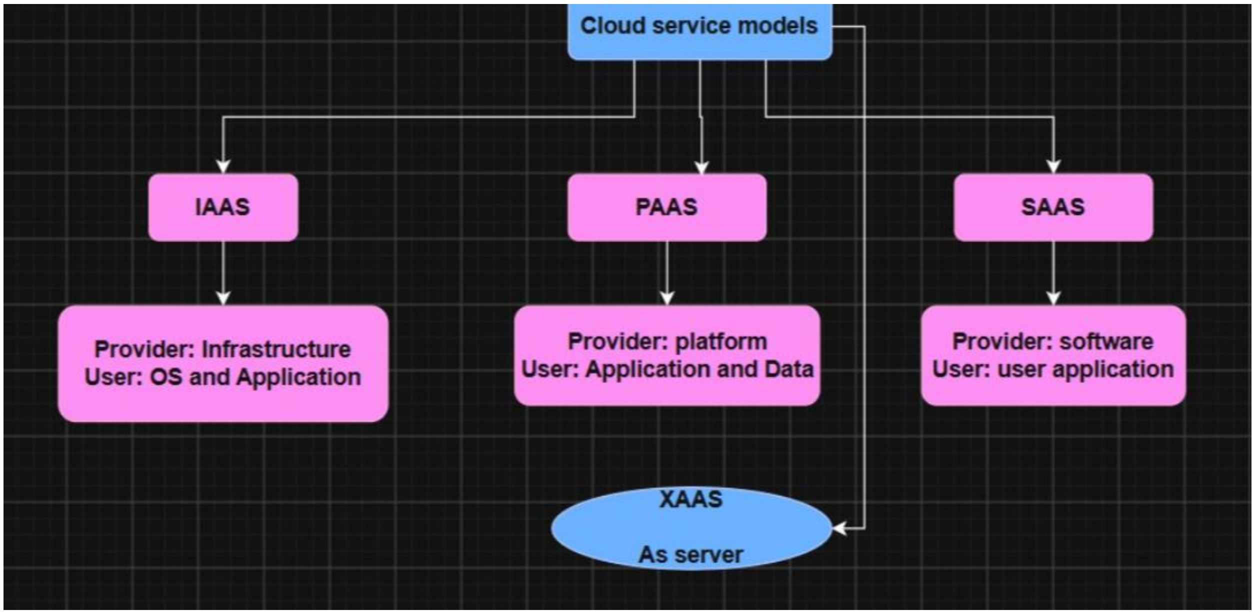
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



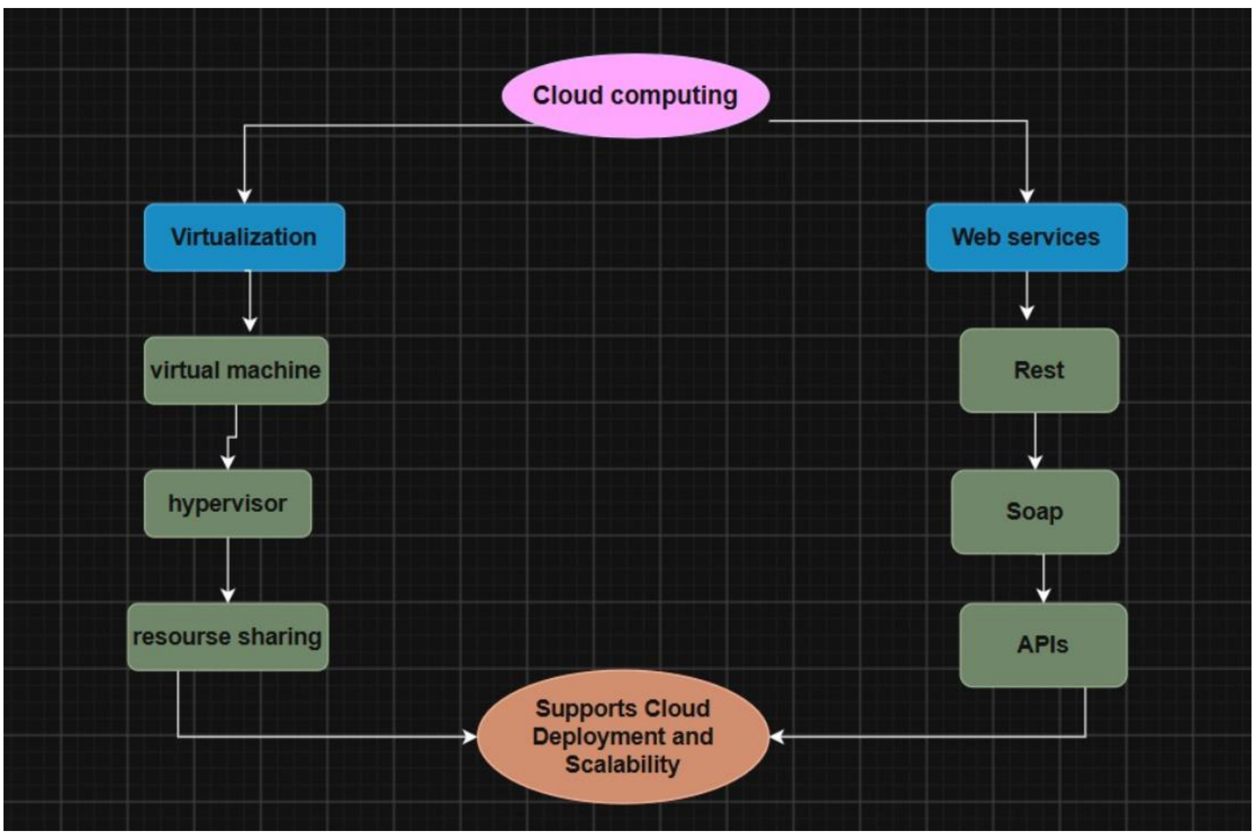
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



5. Questions

i) Evaluate the effectiveness of your concept map in representing cloud computing principles.
The concept map effectively presents the fundamental concepts of cloud computing and illustrates how they are interconnected. Its organized and hierarchical layout helps simplify complex ideas, making the overall architecture easier to understand.

ii) Analyze how scalability, elasticity, and resource management influence the architecture. What happens if they are absent?
Scalability enables the system to handle increasing workloads, while elasticity allows resources to expand or shrink according to demand. Resource management ensures that computing resources are allocated and utilized efficiently. Without these features, the cloud system may experience reduced performance, inefficient resource utilization, increased operational costs, and possible service interruptions.

iii) Justify the selection of the APIs, web services, or cloud services included in your concept map.
The selected APIs and web services facilitate smooth communication and data exchange between different applications. Cloud service models such as IaaS, PaaS, and SaaS were included because they provide essential infrastructure, development platforms, and software solutions, making cloud computing more flexible, scalable, and cost-effective.

iv) Critically examine the relationships between the components in your concept map.
Each component in the concept map is interconnected and contributes to the overall functionality of the cloud environment. Their coordinated interaction supports efficient resource utilization, reliable service delivery, enhanced scalability, and improved user satisfaction.

v) Reflect on the process of developing your concept map. What improvements would you make?
Developing the concept map strengthened my understanding of cloud computing concepts and how different components work together within the architecture. To improve it further, I would include additional elements such as deployment models, security mechanisms, and real-world examples to provide a more comprehensive representation.